Predicting Showjumping Outcomes with Self-Supervised Video Representations: Type-Conditioned Approach Quality from Unlabeled Footage

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Abstract

Whether a showjumping horse clears a fence or knocks a rail is a motion-analysis problem, but predicting it from broadcast video risks learning which arena a clip came from rather than how the horse approached. We study type-conditioned approach quality at a fixed 1.50 m height, using a self-supervised R(2+1)D-18 video encoder pretrained on unlabeled clips and fine-tuned with small heads. Our central finding from earlier work was that the encoder's dominant axis of variation was source venue instead of approach quality. We attack this with a venue-balanced sampler and a domain-adversarial objective intended to suppress venue identity, and we evaluate under an honest leave-one-venue-out protocol whose folds never share a backdrop between train and test. Our central finding is that although the adversarial domain loss converges to chance, a linear probe still recovers venue at 85%, and a t-SNE that looks fully debiased disagrees with the probe; a converged adversarial loss is not invariance. This residual venue signal explains that cross-venue outcome F1 sits at the single-class floor, and the proposed SSL label-efficiency advantage, though present as a low-budget trend, is within seed noise.

1. Introduction

Whether a horse jumps clean or knocks a rail is decided by motion: stride rhythm, approach speed, and the takeoff distance d determine the outcome. Predicting outcome directly from video, however, conflates how well the horse is approaching and how hard the fence is. A vertical and an oxer of the same height pose different problems, so a model trained on mixed fences can shortcut by learning fence type instead of approach quality. We fix the height at 1.50 m and study approach quality conditioned on obstacle type (vertical vs. oxer), taking takeoff distance d as a geometric target recovered from video.

Motivation Per-jump outcome labels are expensive and require domain expertise, while unlabeled 1.50 m footage is abundant publicly. This is exactly the regime where self-supervised learning (SSL) should be used. We pretrain a video encoder on unlabeled clips, then fine-tune small heads on a modest labeled set. We test the hypothesis that SSL-pretrained representations match from-scratch supervised accuracy with half the labels, and beat ImageNet pretraining with fewer still. Beyond equestrian sport, the setting is a clean example of learning behavior representations from broadcast video where the recording environment is strongly correlated with, but causally irrelevant to, the label. That correlation is the central difficulty of this project. In our earlier pipeline, t-SNE of the encoder embeddings clustered cleanly by source broadcast — arena, lighting, footing, and crowd — rather than by approach quality. Because a stratified split lets the same venue appear in both training and test, a model can score well by memorizing a venue. We therefore treat venue as a confound to be measured and removed.

Contributions

1. A diagnosis of domain-adversarial debiasing on broadcast video: the DANN domain loss converges to chance, yet a linear probe still recovers venue at 85% and a t-SNE looks fully debiased. A converged adversarial loss is not invariance, and is only revealed by a direct probe.
2. A leave-one-venue-out protocol (each venue is a distinct broadcast, so the held-out arena is wholly unseen) plus the venue probe and sample-efficiency study that surface the above; under it, the proposed SSL label-efficiency gain is at best a within-noise low-budget trend.
3. An end-to-end pipeline from raw broadcasts to (type, d, outcome) labels (a geometric takeoff-distance estimator calibrated by the standard 3.5 m pole, knockdown mining, and a cross-venue labeled set).

2. Related Work

Self-supervised video representation learning Contrastive instance discrimination (SimCLR [2], MoCo [4]) pulls augmented views of a sample together and pushes other samples apart via the InfoNCE loss [14]. Because appearance-only contrast can ignore motion, video methods add temporal pretext tasks (order verification [10], clip-order prediction [17], pace prediction [16]). We combine InfoNCE with a 6-way temporal-order task on an R(2+1)D-18 backbone [13]. Unlike these methods, evaluated on action-recognition benchmarks where domain is balanced across classes, our footage couples domain (venue) tightly to the recording, so contrastive negatives are venue-dominated unless explicitly controlled.

Domain shift and adversarial debiasing Domain-adversarial training [3] uses a gradient-reversal layer to make features uninformative about domain while staying predictive of the task. Normally applied to cross-dataset adaptation, we repurpose "domain" as the source broadcast within a single task, stripping venue cues (arena, crowd, graphics) that would otherwise act as an outcome shortcut, an instance of the broader problem that vision models exploit backgrounds rather than the object of interest.

Equestrian motion analysis Computer vision for sports centers on tracking, pose, and event detection [12]. Quantitative equine-jumping analysis is mature but relies on self-collected, controlled data: Santamaría et al. [11] track skin markers to measure jumping technique (including the fence-base-to-limb take-off distance we also estimate) and Barrey and Galloux [1] fit an accelerometer belt to eight horses; markerless DeepLabCut [8] pipelines and horse datasets such as Horse-10 [9] and the marker-validated PFERD [7] (5 horses, ten cameras, including jumping) now supply ground-truth kinematics. All are similar in that they examine a few horses recorded by the authors with fixed cameras or worn sensors. We instead learn from pre-existing broadcast footage (three Grand Prix competitions). Uncontrolled video carries a venue signal controlled capture never exposes. To our knowledge this is the first work to learn self-supervised representations of jumping from public competition video, recover takeoff distance geometrically from broadcast frames, and predict jump outcome under an explicit cross-venue protocol. We use YOLOv8 [5] for detection and t-SNE [15] for embedding visualization.

3. Methodology

3.1. Data and Preprocessing

Footage and segmentation We scraped one full 1.50 m Grand Prix broadcast from each of three venues (Tryon International, Club de Campo Villa de Madrid, and Wellington International) and cut them into 2-second clips at 16 fps, 320×320. A YOLOv8 [5] horse-presence filter (keep clips with a horse visible in $\geq 40\%$ of sampled frames) removes crowd shots, course walks, slow-motion replays, and interviews, leaving 2,142 candidate jumping clips, one broadcast per venue. An earlier version of the corpus drew four of its six broadcasts from Wellington, over-representing that single arena and skewing both the venue-balanced sampler and the domain-adversarial head toward it; we deliberately trimmed the set to one broadcast per venue so the three venues contribute comparable footage and leave-one-venue-out is symmetric across them. The trade-off is that, with one competition per venue, venue, competition, course, and camera setup are confounding.

Outcome labels and knockdown mining Clean jumps dominate broadcasts, so naive sampling yields almost no negatives. At demo day the labeled set held 110 outcomes (101 "clean," 9 "knockdown") and the classifier collapsed to always predicting "clean" (macro-F1 ≈ 0.47, the single-class baseline). We therefore mined knockdowns by scrubbing to replay/commentary-flagged rails and cutting tight clips around the fault, growing the set to 346 clips across all three venues — 256 clean and 90 knockdown (2.84:1; madrid 118/15, tryon 65/40, wellington 73/35) — a $3\times$ scale-up over the demo-day 110 that also sharply improves balance. Because refusals are too rare to label reliably at this height, the outcome task reduces to binary clean-vs-knockdown (a deviation from the proposed three-class target; see §5). The set is still imbalanced, so we report macro-F1 as the headline metric.

Geometric takeoff distance Pretrained COCO YOLOv8 (the `horse` class) detects the horse every other frame. The takeoff frame is the sharpest downward-to-upward reversal of the horse-box bottom edge in the clip's second half, a proxy for the hoof leaving the ground. On the subset where the takeoff actually falls inside the clip window (n=27) the detector localizes it to a mean absolute error of 3.3 frames (67% within ± 2); across all 77 annotated clips the error rises to 7.0 frames because roughly two-thirds have the takeoff cut at the segment boundary — a footage/segmentation limit. The standard 3.5 m top pole gives meters-per-pixel $\text{mpp} = 3.5/w_{\text{fence}}$, and d is the ground distance from the takeoff-frame hoof line to the fence base, scaled by mpp (Fig. 1). The fence boxes are 338 hand-annotated rectangles across the three videos, typed by pole count (215 single-plane verticals, 123 two-pole oxers); we additionally fine-tuned a YOLOv8 fence detector on them to recover the proposal's automated detection path.

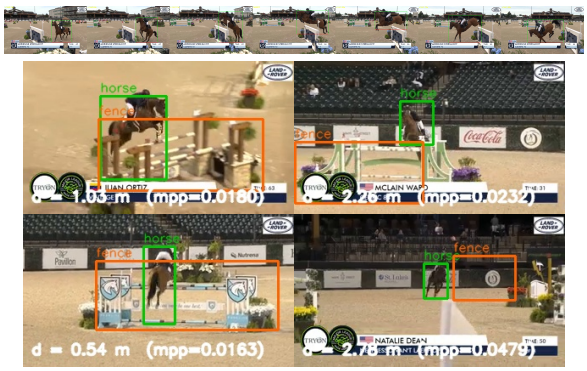


Figure 1. **Detection and geometry.** *Top:* takeoff-frame localization (box-edge reversal, approach→suspension→landing); MAE 3.3 frames (67% within ±2) on the n=27 in-window clips (see text). *Bottom:* horse box (green), fence box (orange), and geometric d at the detected takeoff frame; bottom-right is the oblique-angle failure mode that inflates d.

3.2. Self-Supervised Encoder

The encoder is a torchvision R(2+1)D-18 trunk [13] with a 2-layer MLP projector to 128 dimensions. Three objectives are trained jointly:

- **InfoNCE** [14] on two photometric- and flip-augmented views of a clip (τ=0.1), with the rest of the batch as negatives.
- **6-way temporal-order** classification of the permutation applied to three disjoint 4-frame sub-clips, weight λ_{order}=0.3.
- **Venue-adversarial** prediction of source venue behind a gradient-reversal layer [3], weight λ_{domain}=0.3, pushing the encoder to discard venue-identifying cues.

We optimize with AdamW (lr 10^{-3}, weight decay 10^{-4}, cosine schedule), batch 32, from Kinetics-400 [6] initialization. From-scratch SSL collapsed (nearest-neighbor cosine ≈ 0.97) at this data scale. A venue-balanced sampler equalizes venue representation per batch so contrastive negatives are not dominated by a single broadcast.

3.3. Downstream Heads and Baselines

On the frozen or fine-tuned encoder, we train a type-conditioned outcome classifier and a d regressor. We compare against three baselines: from-scratch supervised R(2+1)D-18, an ImageNet-pretrained 2D-CNN, and a type-only logistic regression. The last shows that if fence type alone predicts outcome well then the task is not measuring approach quality.

3.4. Honest Cross-Venue Evaluation

A stratified split leaks venue; because each venue is a single broadcast, the same arena, crowd, and on-screen graphics would appear in both train and test, letting a model score well by memorizing venue characteristics. With one competition per venue, venue, course, and backdrop are perfectly confounded, so we cannot construct a same-venue/different-competition control. We therefore evaluate strictly leave-one-venue-out: hold out one venue's clips entirely, train on the other two, and repeat for all three folds. The held-out arena is never seen in training, making this a strict test of cross-venue generalization. We quantify residual venue leakage directly with a venue linear probe on frozen embeddings (chance = 13). Sample efficiency is reported under the same held-out-venue split at $n \in \{25, 50, 100\}$ with three seeds, so the SSL-vs-baselines comparison is never inflated by venue leakage.

4. Results and Analysis

Our strongest and most reproducible finding is a negative one about the venue confound; the proposal's headline sample-efficiency comparison, by contrast, is inconclusive at this data scale. We present them in that order.

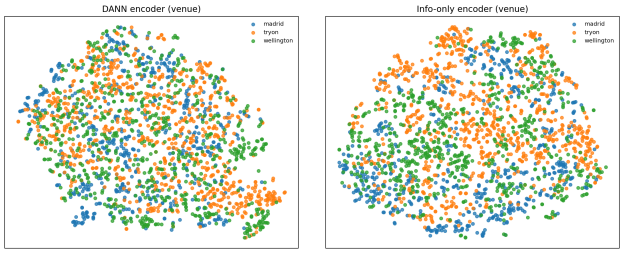
4.1. The venue confound survives adversarial debiasing

This is the paper's central result. During pretraining the adversarial domain loss converged to its three-class chance floor ($L_d \approx \log 3 \approx 1.10$). The venue head could no longer tell broadcasts apart, which is what a successful DANN looks like from the inside. Yet a post-hoc linear probe trained on the frozen embeddings recovers venue at 0.845 ± 0.028 (chance 0.333, majority 0.389; n=2142). The features retain venue information such that the DANN objective was satisfied without achieving invariance.

The t-SNE in Fig. 2 sharpens this point. Both the DANN encoder and an info-only (λ_{domain}=0) ablation are heavily venue-mixed in 2D (an improvement from the milestone), so the projection alone would read as "confound removed." It is not: the same DANN embeddings are 85% venue-decodable. The 2D appearance and the linear decodability give different results, but the probe is quite decisive.

4.2. Cross-venue outcome prediction is at the floor

Because each venue is a single broadcast, there is no within-venue control; we evaluate strictly leave-one-venue-out. Test A (Fig. 3) yields mean macro-F1 0.46 (tryon 0.52, wellington 0.45, madrid 0.43) at mean accuracy 0.58. This sits at the single-class floor (∼0.47), with only tryon clearly above it. Approach-quality signal does not transfer to an unseen venue at three venues, directly consistent with §4.1. If the embeddings still encode venue, a held-out venue is effectively out of distribution. The fold-to-fold spread is large, reflecting how strongly each held-out venue's class balance and appearance dominate at this scale.



(a) DANN encoder. (b) Info-only encoder.

Figure 2. **Embeddings colored by venue:** DANN (left) vs. info-only (right). Both look venue-mixed in 2D, so the projection alone reads as debiased, yet the probe recovers venue at 0.845 from the DANN features. Appearance and decodability are unagreeable but the probe is decisive.

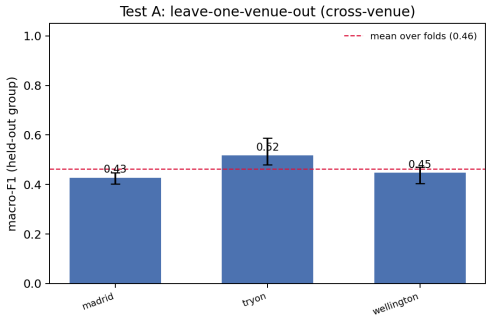


Figure 3. **Leave-one-venue-out (Test A).** Per-fold macro-F1 (mean, min/max over seeds): tryon 0.52, wellington 0.45, madrid 0.43; mean 0.46. Only tryon clears the ∼0.47 single-class floor; with three folds, variance is high.

4.3. Sample efficiency: inconclusive

The proposal's central quantitative test is whether SSL cuts the label budget. Our result is weak. Under a rotating held-out venue (Fig. 4, Table 1) every method sits in a narrow 0.35–0.56 macro-F1 band around the ∼0.47 single-class floor, and the seed-to-seed spread dwarfs the between-method gaps. SSL at n=25 alone spans 0.44–0.61 across its three seeds. With three folds and $n_{\text{val}} \approx 107$, these differences are not statistically separable. The most we can say is that SSL is never the worst method and trends best in the low-label regime; we do not claim a clean ranking or a specific "$N\times$ label saving." The ordering is also venue-sensitive (on a fixed madrid hold-out, (§12b) ImageNet-2D leads instead).

4.4. The minority class is barely detected

The task's real target is near-undetected at this scale. On the confusion fold (Fig. 5, n_{val}=27, two knockdowns) the classifier predicts both knockdowns as clean – knockdown recall 0, i.e. it collapses to a near-clean predictor. The larger held-out-venue sweep shows weak but non-zero knockdown signal (macro-F1 above the floor on some folds), but no evaluation shows reliable knockdown detection. We report macro-F1 rather than accuracy precisely because accuracy hides this; the binary clean-vs-knockdown task is only weakly learned.

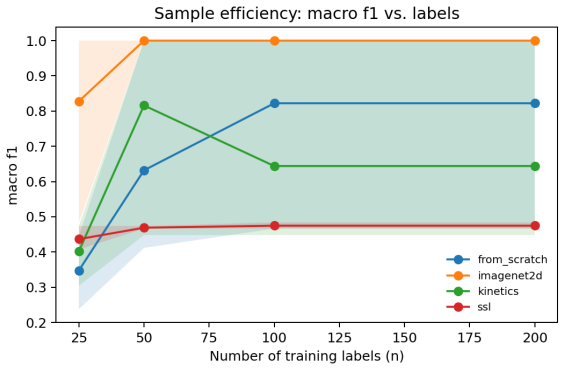


Figure 4. **Sample efficiency under a held-out venue.** Macro-F1 vs. labeled clips $n \in \{25, 50, 100, 200\}$ (rotating held-out venue). All methods overlap around the single-class floor (dashed); between-method differences are within seed variance.

Method (held-out venue)	n=25	n=50	n=100	n=200
SSL (ours)	0.51	0.50	0.54	0.56
ImageNet-2D CNN	0.36	0.53	0.50	0.54
Kinetics-only	0.43	0.37	0.45	0.52
From-scratch supervised	0.35	0.49	0.39	0.48

Table 1. **Sample-efficiency macro-F1** (seed means, rotating held-out venue). All values sit within ∼0.1 of the 0.47 single-class floor; differences are within seed spread and should not be read as a ranking (see §Sample efficiency).

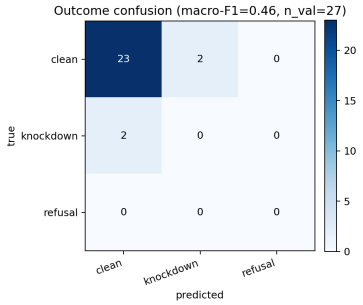


Figure 5. **Confusion matrix**, fine-tuned SSL classifier (macro-F1 0.46, n_{val}=27). Both knockdowns are predicted clean — knockdown recall 0 on this fold. Minority detection is unreliable; the larger sweep (Fig. 4) is the more trustworthy signal.

Geometric d The geometric estimator yields physically plausible takeoff distances (1–3 m; Fig. 1). Learned d-

regression, however, is not useful. Across the sweep the regressor reaches a d-MAE of $\approx$1.2 m on that 1–3 m range, comparable to or worse than simply predicting the mean, so we report it as a negative result and defer a usable regressor to future work.

4.5. Discussion and Limitations

The venue confound. Every downstream weakness above traces back to §4.1. With venue still 85% decodable, an unseen venue is out of distribution, so cross-venue F1 sits at the floor and label-efficiency differences vanish into noise. Stronger invariance (a larger venue penalty, a deeper adversary, or more than three venues so the adversary cannot memorize) is the obvious next step.

Type conditioning gives no measurable benefit. The title frames type-conditioned approach quality, but a held-out-venue ablation ($n_{\text{val}} \approx 107$, 3 seeds) does not support the conditioning. The type-conditioned head reaches macro-F1 0.55 versus 0.58 for the unconditioned one, if anything slightly worse, and within seed noise either way. Type conditioning is thus a modeling choice our data does not validate. Whether it helps at larger scale, or with a learned rather than pole-count fence type, is uncertain.

Class imbalance and small folds. With a clean-dominated, refusal-free set, accuracy is misleading and macro-F1 is most important. Leave-one-venue-out gives one venue per fold ($n_{\text{val}} =$ madrid 133, tryon 105, wellington 108) and the confusion fold is smaller (27); three folds and three seeds remain noisy, which is itself a limitation on every claim above.

Geometric d depends on a fragile takeoff frame. The pole-scaled estimator inflates d when the camera is off-axis (Fig. 1), and d inherits the takeoff-frame error: reliable only on the $\sim$one-third of clips whose takeoff falls inside the 2-second window (MAE 3.3 frames), and unreliable on the rest, where it is cut at the boundary. Side-view filtering and wider windows would trade pool size for accuracy.

5. Conclusion

We studied type-conditioned showjumping outcome prediction as a self-supervised video-representation problem whose central obstacle is a venue confound such that broadcast environment correlates strongly with, but is causally irrelevant to, the label. We attacked it with a venue-balanced sampler and a domain-adversarial objective, and evaluated under a strict leave-one-venue-out protocol. The most important finding is that the domain-adversarial debiasing did not achieve invariance, although its loss converged to chance, venue stayed $\sim$85% linearly decodable, and a t-SNE that looks debiased disagrees with the probe. Cross-venue outcome F1 sits at the single-class floor, the proposal's SSL label-efficiency advantage appears only as a low-budget trend within seed noise rather than a separable win, and the minority class (rails down) is only weakly detected at this scale. This confound caps how far any of this work transfers to accuracy.

Future work Larger and more balanced knockdown labels, recovery of the three-class outcome (including refusals), isolating the individual pretext tasks (InfoNCE-only vs. temporal-order-only, which we did not run, only the type-conditioning ablation, which showed no benefit), reinstating d-regression with side-view filtering, and the deferred future-frame prediction task once the confound is controlled.

Key takeaway A converged adversarial loss is not invariance. Our venue head reached its chance floor while a linear probe still read venue at 85%. Evaluating self-supervised video representations under domain shift therefore demands both a leakage-free split and a direct probe of the nuisance variable.

6. Individual Contributions

This is a solo project. I designed and implemented the full pipeline (scraping, segmentation, horse/fence detection, geometric d, knockdown mining, SSL pretraining with the venue-adversarial head, downstream heads, baselines, and the cross-venue evaluation), and ran all experiments. Claude Code was used to help write the in-Colab annotator and scraping script, evaluation/visualization scaffolding, and code documentation. Codebase: https://github.com/bballhaus/showjumping-ssl.

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